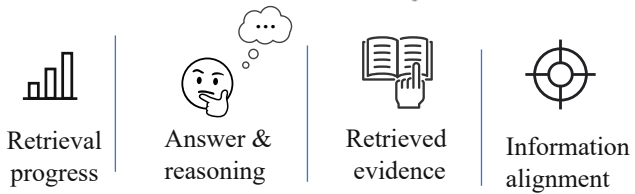


## Pre-action state

State representation  $z_i^{(r)} \in \mathbb{R}^{32}$



Which query should receive the next retrieval opportunity?

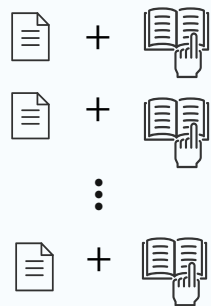
$\phi(\cdot)$

Queries

$q_1$   
 $q_2$   
 $\vdots$   
 $q_n$



Answer + evidence



## Single-step retrieval value

Pre-action state

$z_i^{(r)} \in \mathbb{R}^{32}$



What is the value of the next retrieval?

Value estimator

$f_\theta(\cdot)$



value

$\hat{\eta}_i^{(r)}$

train  $f_\theta$

Offline training supervision

$Q_i^{(r)}$



Retrieve once



$Q_i^{(r+1)}$



$$\Delta_i^{(r+1)} = Q_i^{(r+1)} - Q_i^{(r)}$$

## Sequential shared-budget allocation

Shared budget B



1 budget unit

Current action pool

(query, action)  $\hat{\eta}_i^{(r)}$

$(q_7, a_7^{(1)}) + 0.08$   
 $(q_4, a_4^{(1)}) + 0.06$   
 $(q_9, a_9^{(2)}) + 0.04$   
 $\vdots$   
 $(q_2, a_2^{(1)}) - 0.01$

Next allocation step

Updated action pool

(query, action)  $\hat{\eta}_i^{(r)}$

next  $(q_4, a_4^{(1)}) + 0.06$   
 $(q_9, a_9^{(2)}) + 0.04$   
 $(q_7, a_7^{(2)}) + 0.03$   
 $\vdots$   
 $(q_2, a_2^{(1)}) - 0.01$

One allocation step

Retrieve once



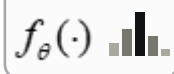
Update state

$z^{(r)} \rightarrow z^{(r+1)}$

Next legal action?

Yes

Re-estimate



Reinsert



No

Exit pool